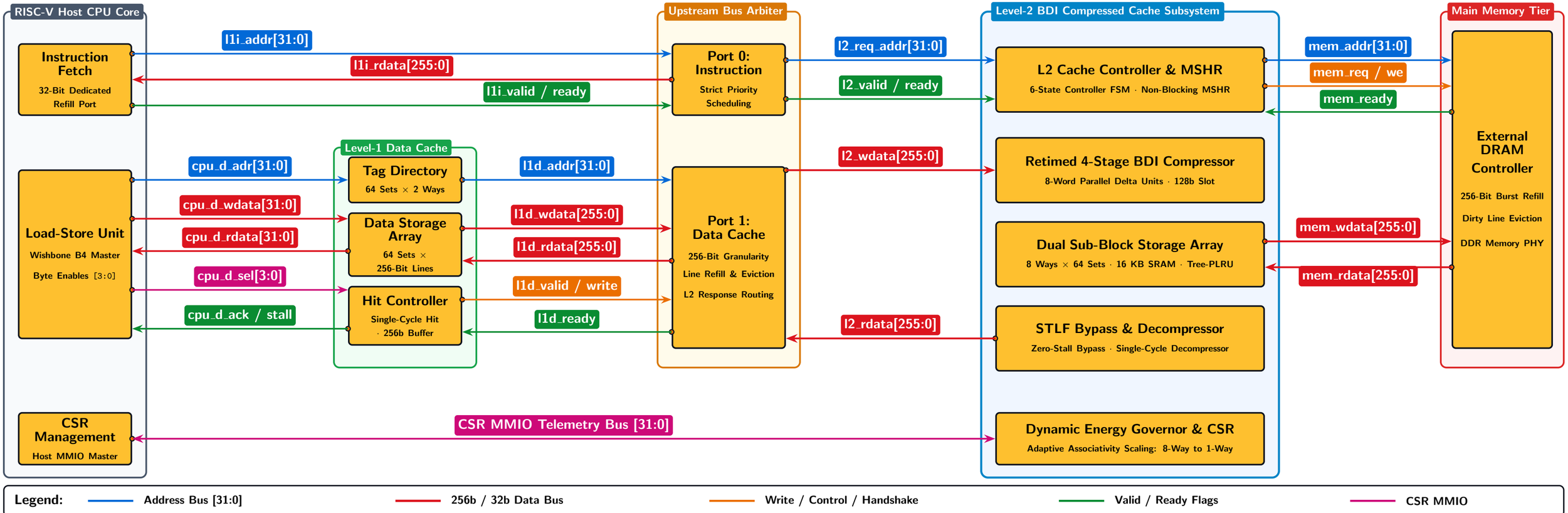
RISC-V Host CPU Core
Instruction Fetch
32-Bit Dedicated Refill Port
Load-Store Unit
Wishbone B4 Master
Byte Enables [3:0]
CSR Management
Host MMIO Master
Level-1 Data Cache
Tag Directory
64 Sets × 2 Ways
Data Storage Array
64 Sets × 256-Bit Lines
Hit Controller
Single-Cycle Hit · 256b Buffer
Upstream Bus Arbiter
Port 0: Instruction
Strict Priority Scheduling
Port 1: Data Cache
256-Bit Granularity
Line Refill & Eviction
L2 Response Routing
Level-2 BDI Compressed Cache Subsystem
L2 Cache Controller & MSHR
6-State Controller FSM · Non-Blocking MSHR
Retimed 4-Stage BDI Compressor
8-Word Parallel Delta Units · 128b Slot
Dual Sub-Block Storage Array
8 Ways × 64 Sets · 16 KB SRAM · Tree-PLRU
STLF Bypass & Decompressor
Zero-Stall Bypass · Single-Cycle Decompressor
Dynamic Energy Governor & CSR
Adaptive Associativity Scaling: 8-Way to 1-Way
Main Memory Tier
External DRAM Controller
256-Bit Burst Refill
Dirty Line Eviction
DDR Memory PHY
I1i_addr[31:0]
I1i_rdata[255:0]
I1i_valid / ready
cpu_d_adr[31:0]
cpu_d_wdata[31:0]
cpu_d_rdata[31:0]
cpu_d_sel[3:0]
cpu_d_ack / stall
I1d_addr[31:0]
I1d_wdata[255:0]
I1d_rdata[255:0]
I1d_valid / write
I1d_ready
I2_req_addr[31:0]
I2_valid / ready
I2_wdata[255:0]
I2_rdata[255:0]
mem_addr[31:0]
mem_req / we
mem_ready
mem_wdata[255:0]
mem_rdata[255:0]
CSR MMIO Telemetry Bus [31:0]
Legend:
Address Bus [31:0]
256b / 32b Data Bus
Write / Control / Handshake
Valid / Ready Flags
CSR MMIO